

Laravel / PHP / MySQL Developer Assignment

1. Project Title

Mini Order Management API

Build a small backend system where users can create products and place orders using REST APIs.

2. Objective

Evaluate Laravel fundamentals, PHP coding practices, MySQL database design, API development, and problem-solving ability.

3. Technical Requirements

Laravel 10+

PHP 8+

MySQL

REST API

Git repository

Optional Bonus:

Docker

Unit testing

API documentation

4. Features to Implement

4.1 User Authentication

Implement API authentication.

Required endpoints:

- Register user
- Login user

- Logout user

Suggested: Laravel Sanctum or JWT

4.2 Product Management

Create CRUD APIs for products.

APIs:

- Create product
- List products
- Update product
- Delete product
- Get single product

4.3 Order System

APIs:

- Create order
- Get user orders
- Get order details

4.4 Business Logic

When placing an order:

1. Check product stock
2. Reduce stock
3. Calculate total price
4. Save order items

Return proper JSON responses.

5. R&D Task

- Redis caching for products
- API rate limiting
- Queue for order processing
- Email notification after order

- Product search filters

Explain the implementation in README.

6. Database Requirements

Use foreign keys

Use Laravel migrations

Use seeders for sample data

7. Code Quality

MVC structure

Clean code

Proper validation

Error handling

Use Laravel features (Eloquent, Requests, Resources)

8. Submission Requirements

GitHub repository

README with setup instructions

API endpoint list

Explanation of R&D feature

9. Bonus Points

Docker setup

PHPUnit tests

Swagger / OpenAPI documentation

Postman collection

10. Expected Time

Estimated completion time: 18-24 hours